

OKMULGEE, OK, USA

WMO: 722164

Lat: 35.668N

Lon: 95.949W

Elev: 220

StdP: 98.72

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53949

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.5	(c) -7.2	(d) -15.5	(e) 1.0	(f) -7.3	(g) -12.7	(h) 1.3	(i) -3.6	(j) 10.9	(k) 11.9	(l) 10.1	(m) 12.0	(n) 3.0	(o) 10	(p) 0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	38.6	25.0	37.1	24.9	35.0	24.9	27.3	33.9	26.5	32.9	25.8	32.1	4.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	22.0	30.4	25.0	20.6	29.3	24.0	19.4	28.5	87.2	33.7	84.1	32.9	81.0	32.4	36.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.1	(b) 8.8	(c) 8.0	(e) -13.9	(f) 41.0	(g) 3.2	(h) 3.7	(i) -16.2	(j) 43.6	(k) -18.1	(l) 45.8	(m) -19.9	(n) 47.9	(o) -22.2	(p) 50.6		
(4) 10.1	(5) 8.8	(6) 8.0	(7) -14.3	(8) 28.4	(9) 3.0	(10) 2.6	(11) -16.5	(12) 30.3	(13) -18.2	(14) 31.9	(15) -19.9	(16) 33.4	(17) -22.0	(18) 35.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	4.8	6.4	12.1	16.5	20.6	25.7	27.8	27.6	23.4	16.7	11.1	5.4		
	DBStd	9.54	6.00	6.39	5.78	4.96	4.10	2.94	3.06	3.37	3.75	5.10	5.44	5.55		
	HDD10.0	586	181	134	43	8	1	0	0	0	0	7	51	161		
	HDD18.3	1802	421	338	206	89	26	0	0	0	5	93	224	400		
	CDD10.0	2979	19	33	107	204	329	472	551	544	402	216	83	20		
	CDD18.3	1155	1	3	12	35	96	223	293	286	157	43	7	1		
	CDH23.3	12827	6	19	93	364	847	2464	3593	3411	1556	414	58	3		
	CDH26.7	6218	0	2	18	98	257	1166	1968	1881	700	121	8	0		
(14) Wind	WSAvg	3.4	3.6	3.8	4.2	4.3	3.6	3.4	2.7	2.7	2.8	3.2	3.6	3.4		
(15) Precipitation	PrecAvg	1035	43	52	87	108	141	111	80	75	105	98	86	61		
	PrecMax	1432	109	142	240	285	260	266	240	302	280	384	231	220		
	PrecMin	544	0	4	0	10	45	19	0	0	0	0	0	3		
	PrecStd	193	29	37	50	59	64	56	68	54	76	82	58	52		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.8	26.3	28.8	33.0	32.8	38.0	41.2	42.3	37.3	32.2	27.6	22.9		
		MCWB	14.2	17.5	18.9	22.8	23.3	26.2	24.9	24.8	23.5	22.8	18.5	16.3		
	2%	DB	21.0	22.7	26.1	28.9	31.2	36.1	38.0	38.9	34.5	29.9	25.0	20.2		
		MCWB	13.3	15.4	17.3	20.2	23.3	24.6	24.9	24.9	24.0	21.8	16.7	14.1		
	5%	DB	17.8	19.8	23.5	27.3	29.3	33.8	37.0	37.2	32.8	27.6	22.5	17.3		
		MCWB	12.4	14.3	16.3	18.9	22.1	24.8	25.0	24.9	23.4	19.5	16.4	12.7		
	10%	DB	15.2	17.2	21.9	25.4	27.8	32.5	34.9	34.9	31.1	25.8	20.1	14.4		
		MCWB	9.5	12.0	15.2	18.3	21.4	24.5	25.3	24.7	22.9	19.0	14.6	10.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	19.1	20.6	23.8	26.0	28.0	28.6	27.9	26.0	24.1	20.9	18.7		
		MCDB	19.5	23.5	26.4	28.9	30.5	34.1	35.0	35.1	31.9	29.4	24.9	22.2		
	2%	WB	15.5	17.5	19.1	22.2	24.6	26.9	27.4	26.9	25.2	23.0	19.1	16.4		
		MCDB	18.9	20.7	23.9	27.1	28.9	33.1	33.5	34.3	31.2	27.9	22.9	18.9		
	5%	WB	12.5	15.3	18.0	20.7	23.7	25.9	26.4	26.2	24.5	21.9	17.4	13.5		
		MCDB	17.0	18.7	21.8	25.1	27.6	31.8	32.4	33.4	30.3	25.3	20.7	15.8		
	10%	WB	10.0	12.3	16.6	19.5	22.7	25.2	25.8	25.5	23.8	20.6	15.6	11.0		
		MCDB	14.6	16.4	20.4	23.7	26.4	31.1	32.0	32.3	29.1	23.9	19.4	13.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	11.7	12.3	12.3	10.9	11.2	12.0	12.3	12.7	12.9	12.7	11.2		
		MCDBR	16.6	16.0	14.5	14.5	12.5	13.3	14.5	15.1	14.8	14.5	15.0	15.4		
	5% WB	MCWBR	10.4	9.8	7.9	7.8	5.9	5.3	4.2	4.3	5.4	6.9	8.8	10.1		
		MCWBR	14.6	14.0	11.6	12.0	10.7	11.9	11.6	12.7	12.1	11.3	12.2	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.309	0.350	0.385	0.421	0.429	0.441	0.441	0.393	0.350	0.319	0.298		
	taud		2.496	2.492	2.407	2.322	2.285	2.292	2.303	2.263	2.368	2.477	2.507	2.517		
	Ebn at Noon		904	937	920	899	867	856	843	836	862	877	875	885		
	Edh at Noon		82	93	110	126	133	132	130	133	114	93	81	76		
(44) All-Sky Solar Radiation	RadAvg		2.60	3.15	4.16	5.18	5.67	6.51	6.60	5.99	4.98	3.75	2.76	2.21		
	RadStd		0.27	0.47	0.26	0.33	0.54	0.67	0.38	0.38	0.43	0.49	0.34	0.28		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page